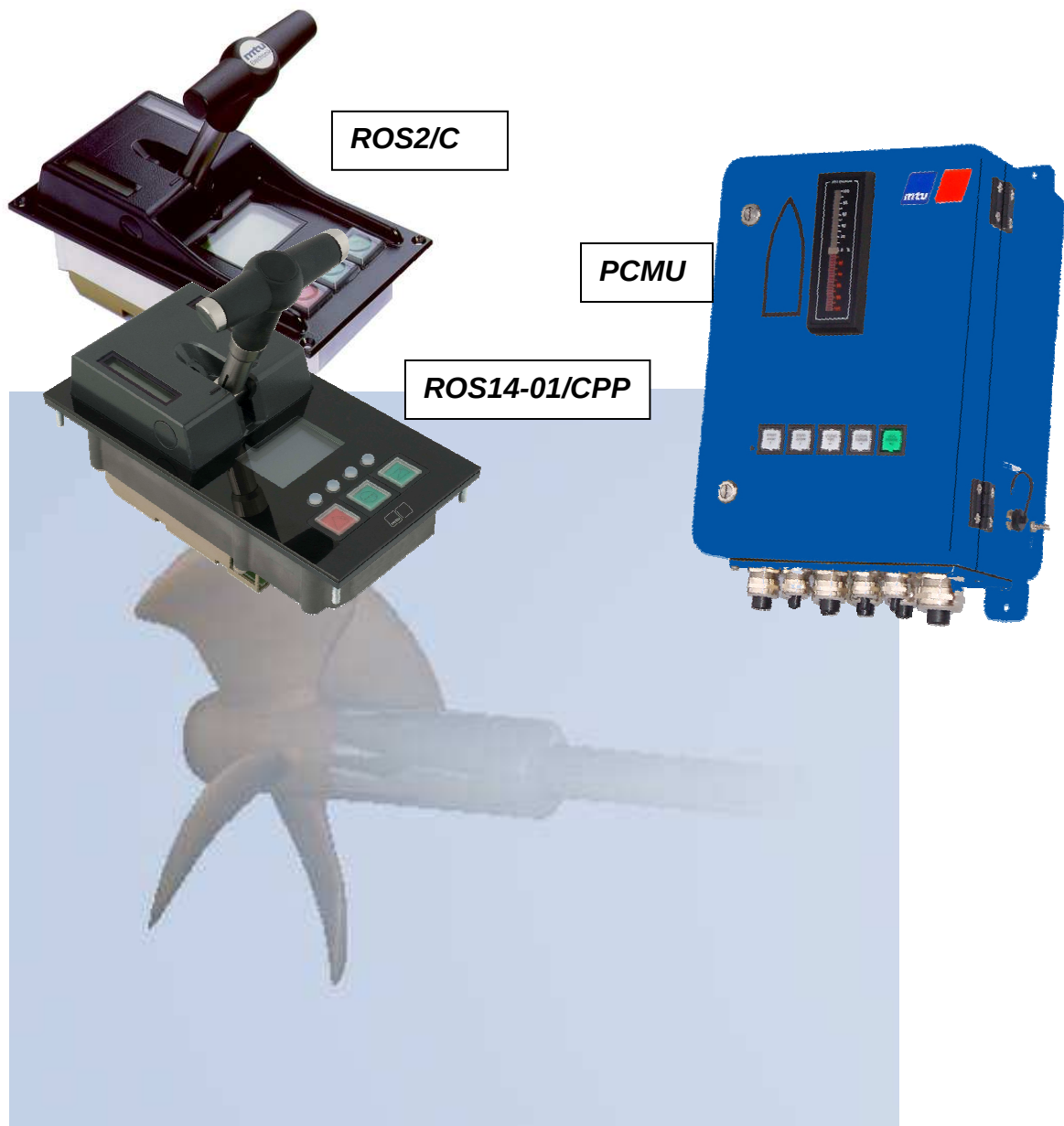


CPP system settings

CCP = controllable pitch propeller-



For commissioning and service through

- Settings in the control of the CPP hydraulic system (PCMU)
- Settings in the RCS
- Settings during SAT



**Work on the engine and automation system
may only be performed by trained and in-
structed professionals.**



CPP system settings

CCP = controllable pitch propeller-

Procedure

1. Making Settings for Pitch Feedback Signals

2. Local Pitch Control with Disengaged Propeller Shaft

3. Setting the Proportional Valve

3.1. Control by MOB2

3.2. Control by -10V/+10V Signal

3.3. Control by 4-20mA Signal

4. Remote Pitch Control with Disengaged Propeller Shaft

5. Programming Propeller Control Curves in accordance with Shipyard's Specifications

6. Programming Pitch Control Rate in accordance with Manufacturer's Specifications

7. Adjusting the Control Loop (PID)

8. Pitch Control with Engaged Propeller shaft

9. Settings during the Sea Acceptance Tests (SAT)

9.1. Soft Start, Run-down Phase

9.2. Dead Band

9.3. Acceleration Control

9.4. Load Control

9.5. Control Voltage Offset (installation-dependent)



CPP system settings

CCP = controllable pitch propeller-

1. Making Settings for Pitch Feedback Signals

The feedback sensors must have been adjusted mechanically and electronically by the manufacturer/shipyard in order to be able to take over the feedback signal values.



Sensors 4-20mA, adjustable

- The initial values 4mA and 20mA are supposed to correspond to the mechanical limit stops.

Potentiometer

- The potentiometer value is supposed to be between 2kΩ and 10kΩ.
- The mechanical zero pitch should be equal to half of the potentiometer value.
- Approx. 80% of potentiometer value should be used for the feedback signal.

The redundant sensors are to be adjusted or set to the same value.

Preparatory steps

- Connect dialog unit
- Record values with the following measuring points
 - o **Pitch Position PMU**
 - o **Pitch Position PCU**

Making settings

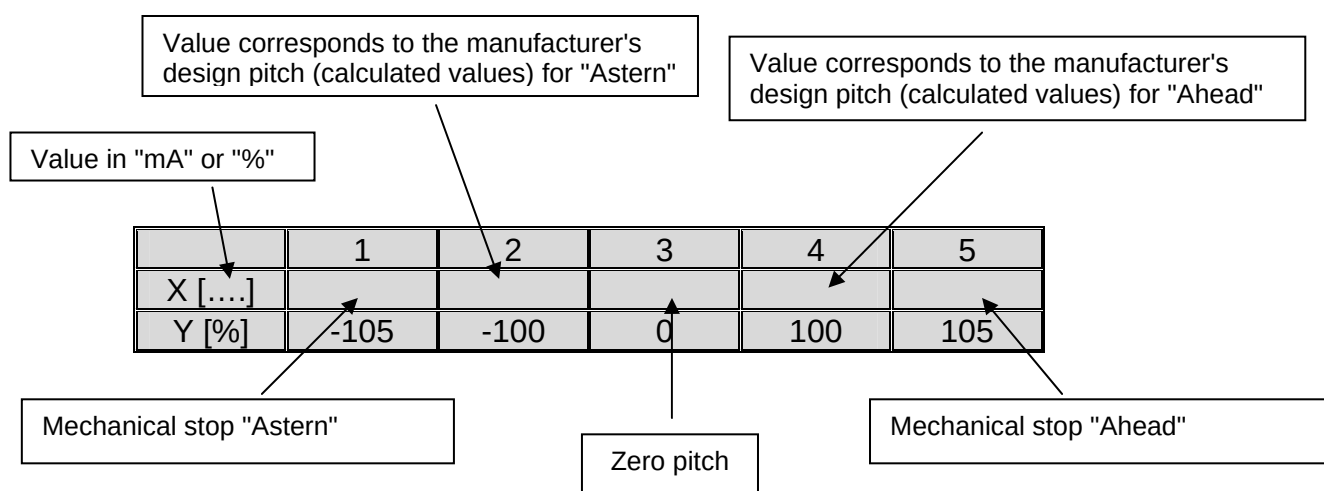
- Disconnect electric control cable of the proportional valves ☐
- Start electric or mechanical hydraulic pump ☐
- Use mechanical pitch control to set pitch to full ahead (mechanical stop) ☐
Take over measured value from **Pitch Position PMU** in curve
Feedback Pitch (PMU) X5 ☐
Take over measured value from **Pitch Position PCU** in curve
Feedback Pitch (PCU) X5 ☐
- Use mechanical pitch control to set pitch to full astern (mechanical stop) ☐
Take over measured value from **Pitch Position PMU** in curve
Feedback Pitch (PMU) X1 ☐
Take over measured value from **Pitch Position PCU** in curve
Feedback Pitch (PCU) X1 ☐



CPP system settings

CCP = controllable pitch propeller-

- Use mechanical pitch control to set pitch to zero ☐
- Take over measured value from **Pitch Position PMU** in curve ☐
- Feedback Pitch (PMU) X3** ☐
- Take over measured value from **Pitch Position PCU** in curve ☐
- Feedback Pitch (PCU) X3** ☐
- Connect electric control cable on the proportional valves ☐



The propeller's 100% pitch is determined during the Sea Acceptance Test and entered on the table in X4 for "Ahead" and in X2 for "Astern". At 100% pitch for "Ahead", the maximum engine speed should be reached at 100% injection quantity. If the pitch is set too high, there is no longer any engine power reserve and load control is activated → continuous control. With these settings, the current vessel load should also be taken into account (possibly include power reserve in the calculation). Zero thrust X3 is set during the HAT according to the ship master's indication.

2. Local Pitch Control with Disengaged Propeller Shaft

Local pitch control using the PCMU control buttons.

- Control direction ahead/astern is correct ☐
- 100% ahead and astern are reached ☐



CPP system settings

CCP = controllable pitch propeller-

3. Setting the Proportional Valve

3.1. Calibrating the Proportional Valve when Controlled by a MOB 2

Preparatory steps

- Connect dialog unit
- Record the following parameters:
 - o **Actual Current Ahead**
 - o **Current Demand Ahead**
 - o **Actual Current Astern**
 - o **Current Demand Astern**

Ahead valve setting

- Disconnect valve connector Yxx Ahead ☐
- Write down the value of the **Actual Current Ahead** measuring points and take over as correction value in parameter **Stromerfassungsoffset B** - **Write down original value !!**

Parameter	Original value	Value
Stromerfassungsoffset B		

- Connect valve connector Yxx Ahead ☐
- Set parameter values **R1 of output amplifier B** and **R2 of output amplifier B** to zero - **Write down original values !!!**

Parameter	Original value	Value
R1 of output amplifier B		0
R2 of output amplifier B		0

- Change all Y values of curve **valve 1 curve** to "500" - **Write down original values!!!**

	1	2	3	4	5
X [%]	0.000	0.200	10.000	15.000	30.000
Y [mA]	0.000				



CPP system settings

CCP = controllable pitch propeller-

Values for settings:

	1	2	3	4	5
X [%]	0.000	0.200	10.000	15.000	30.000
Y [mA]	500.000	500.000	500.000	500.000	500.000

- Change value of parameter **24V 100%PWM Duration Strom B** until the measuring point **Actual current ahead** has reached the value "500".

Parameter	Value
24V 100%PWM Duration Strom B	

- In curve **valve 1 curve**, program the original values for parameters **R1 of output amplifier B** and **R2 of output amplifier B**.

Astern valve setting

- Disconnect valve connector Yxx Astern
- Write down the value of the **Actual Current Astern** measuring points and take over as correction value in parameter **Stromerfassungsoffset D** - **Write down original value !!**

Parameter	Original value	Value
Stromerfassungsoffset D		

- Connect valve connector Yxx Astern
- Set parameter values **R1 of output amplifier D** and **R2 of output amplifier D** to zero - **Write down original values !!!**

Parameter	Original value	Value
R1 of output amplifier D		0
R2 of output amplifier D		0

- Change all Y values of curve **valve 2 curve** to "500" - **Write down original values!!!**
Original values **valve 2 curve**:

	1	2	3	4	5
X [%]	0.000	0.200	10.000	15.000	30.000
Y [mA]	0.000				



CPP system settings

CCP = controllable pitch propeller-

Values for settings:

	1	2	3	4	5
X [%]	0.000	0.200	10.000	15.000	30.000
Y [mA]	500.000	500.000	500.000	500.000	500.000



- Change value of parameter **24V 100%PWM Duration Strom D** until the measuring point **Actual current ahead** has reached the value "500".

Parameter	Value
24V 100%PWM Duration Strom D	



- In curve **valve 2 curve**, program the original values for parameters **R1 of output amplifier D** and **R2 of output amplifier D** .



CPP system settings

CCP = controllable pitch propeller-

3.2. Calibrating the Proportional Valve when Controlled by a +/- 10 V signal

This setting procedure determines the behavior of the proportional valve in response to the control signal (V to %pitch/s). The value of the control signal for the minimum pitch adjustment rate and for the maximum possible pitch adjustment rate (when this value is reached, pitch rate does not increase any more) must be determined.

The control signal value 0V corresponds to no valve control which means no change in pitch angle.



Caution

Offset values (**Control_voltage_offset_ahead**) must be deactivated during calibration

Preparatory steps

- Connect dialog unit
- Record the values of the following measuring points
 - **Pitch Position**
 - **Pitch Actuator Demand +/-10V**

2D parameter - (30) **Pitch Actuator Calibration**

	1	2	3	4	5	6	7	8	9	10
X [%]	-13.000	-10.000	-1.200	-0.200	0.000	0.200	1.200	2.000	10.000	13.000
Y [V]					0.000					

Max. pitch control rate (pointing to X[1] = -13.000)

Min. pitch control rate (pointing to X[6] = 0.200)

Min. pitch control rate (pointing to Y[3] = 0.000)

Max. pitch control rate (pointing to Y[9] = 0.000)



Example

Determination of the values for pitch control

- Set the values **Pitch Actuator Calibration** Y6-10 - Y1-4 for astern - to a value "x".
- Move control lever from NEUTRAL to FULL AHEAD.
- Calculate value based on the Diasys records → $\Delta \text{pitch position} \% / \Delta \text{time}$ is the pitch control rate %/s (values during linear control)
- Record the values in a table.



CPP system settings

CCP = controllable pitch propeller-

The initial value for pitch control in ahead direction is 1.3V. If there is already pitch control for this value, a lower value must be selected and this value must be increased stepwise in increments of 0.1-0.2V. When the pitch control curve becomes linear, the value can be increased to 0.5V. The initial value for pitch control in astern direction is 1.3V. If pitch angle is already out of zero with this value, a lower value must be selected and this value must be increased stepwise in increments of -0.1-0.2V. When the pitch control curve becomes linear, the value can be increased to -0.5V.

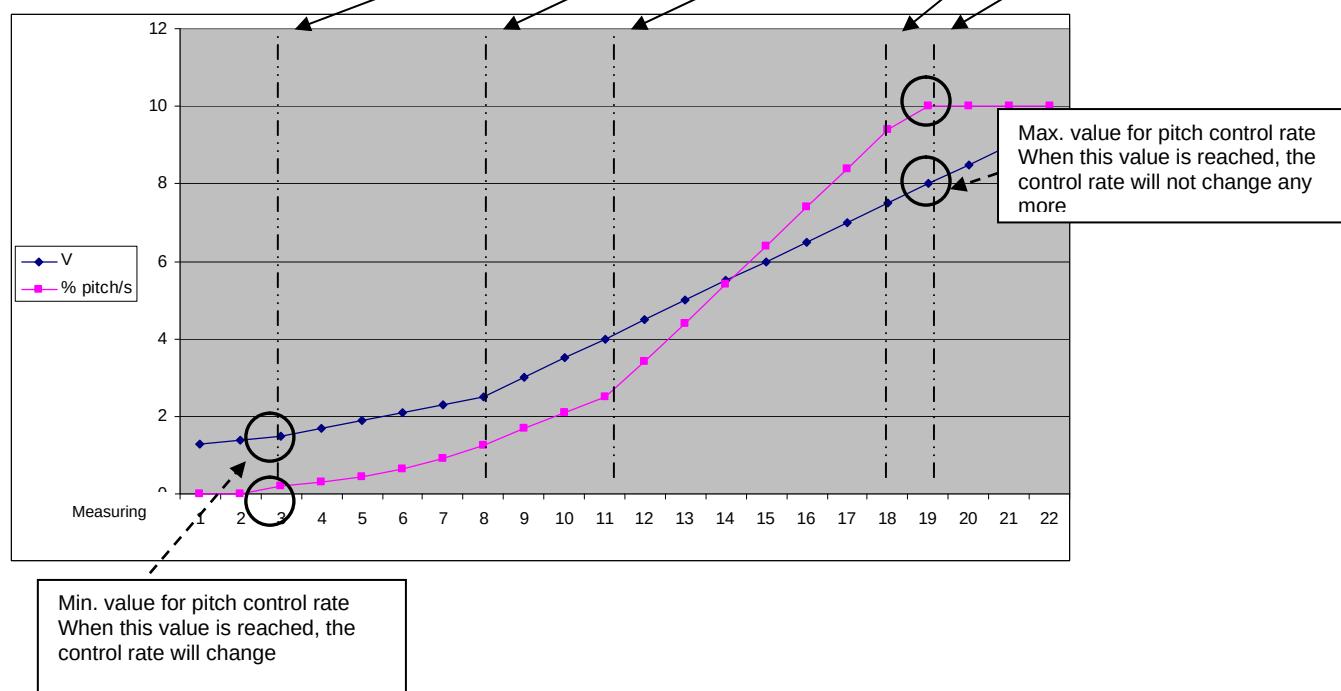
Table and graph derived from it

Worksheet for generating the diagram

Measuring points	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
V	1.3	1.4	1.5	1.7	1.9	2.1	2.3	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5
% pitch/s	0	0	0.2	0.3	0.5	0.7	0.9	1.3	1.7	2.1	2.5	3.4	4.4	5.4	6.4	7.4	8.4	9.4	10	10	10	10

2D parameter - (30) Pitch Actuator Calibration

	1	2	3	4	5	6	7	8	9	10
X [%]	-13.000	-10.000	-1.200	-0.200	0.000	0.200	1.200	2.000	10.000	13.000
Y [V]					0.000	1.500	2.000	3.100	7.000	8.000



CPP system settings

CCP = controllable pitch propeller-

3.3. Calibrating the Proportional Valve when Controlled by a 4-20mA signal

This setting procedure determines the behavior of the proportional valve in response to the control signal (mA to %pitch/s). The value of the control signal for the minimum pitch adjustment rate and for the maximum possible pitch adjustment rate (when this value is reached, pitch rate does not increase any more) must be determined.

The control signal value 12mA corresponds to no valve control which means no change in pitch angle.

Preparatory steps

- Connect dialog unit
- Record the values of the following measuring points
 - o **Pitch Position**
 - o **Pitch Actuator Demand 4-20mA**

2D parameter - **Pitch Actuator Calibration**

	Max. pitch control rate					Min. pitch control rate				
	1	2	3	4	5	6	7	8	9	10
X [%]	-13.000	-10.000	-1.200	-0.200	0.000	0.200	1.200	2.000	10.000	13.000
Y [mA]					12.000					

Min. pitch control rate

Max. pitch control rate



Example

Determination of the values for pitch control

- Set the values **Pitch Actuator Calibration** -Y6-10 - Y1-4 for astern - to a value "x".
- Move control lever from NEUTRAL to FULL AHEAD.
- Calculate value based on the Diasys records → $\Delta \text{pitch position} \% / \Delta \text{time}$ is the pitch control rate %/s (valued during linear control)
- Record the values in a table.

The initial value for pitch control in ahead direction is 10.5 mA. If pitch angle is already out of zero with this value, a higher value must be selected and this value must be reduced stepwise in increments of 0.1mA. When the pitch control curve becomes linear, the value can be increased to 0.5mA. The initial value for pitch control in astern direction is 13.5 mA. If pitch angle is already out of zero with this value, a lower value must be selected and this value must be increased stepwise in increments of 0.1mA. When the pitch control curve becomes linear, the value can be increased to 0.5mA.

